

S
Full Environment State

$$R = \rho(S)_{state \text{ abstraction}}$$

R
Compressed Representation

C_{contract} ContractSoundness

$C_{\text{uncertainty}}$ UncertaintyBound

C_{temporal} TemporalConsistency

$\pi^*(R)$
Decision-Preserving Policy

Each capability is
necessary: removing
any one causes
decision divergence

$$\exists R: \pi_M(R) \neq \pi_{M \setminus \{C\}}(R)$$